

Q9 A and B are correct as they are valid assumptions in the simple kinetic theory of gases, i.e. how ideal gases particles should behave.

C is correct as the total energy of the system must remain constant due to the system being isolated.

D is not correct – given that the initial momentum of a molecule before collision is p , there will be a change in momentum of $2p$ because the final momentum of the molecule will be $-p$ (same magnitude, opposite in direction).

Answer: D

Q10 The pressure of a gas is derived from the forces of collision between the molecules and